TECHNICAL CASE STUDY · ENGLISH EDITION

MADA-BMP

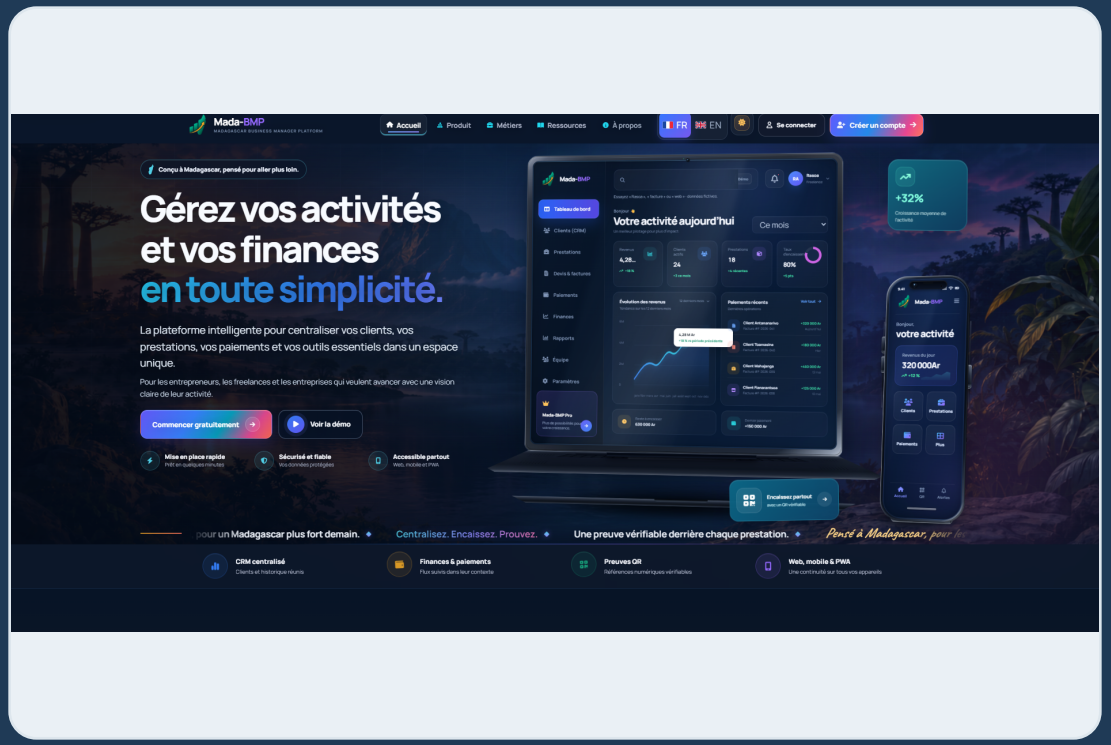
Madagascar Business Manager Platform

MASTER 1 ACADEMIC PROJECT - IN PROGRESS

A web platform for managing professional activities, designed for independent professionals and small teams.

TECHNOLOGIES

React / TypeScript · Spring Boot · FastAPI / Python · PostgreSQL



Illustrative public landing page · demonstration data. 2026.

Presented by: RANDRIAMANANTENA Fitahiana Lafatra

A project to organize small-business operations

Context, objective, target audience and technology stack - without confusing product vision with validated features.

CONTEXT · BUSINESS CHALLENGE

Client records, requests, quotations, invoices and payments are often scattered across notebooks, messages and files.

MISSION · OBJECTIVE

Connect customer relationships, services, documents and payment tracking within a shared workspace.

AUDIENCE · TARGET USERS

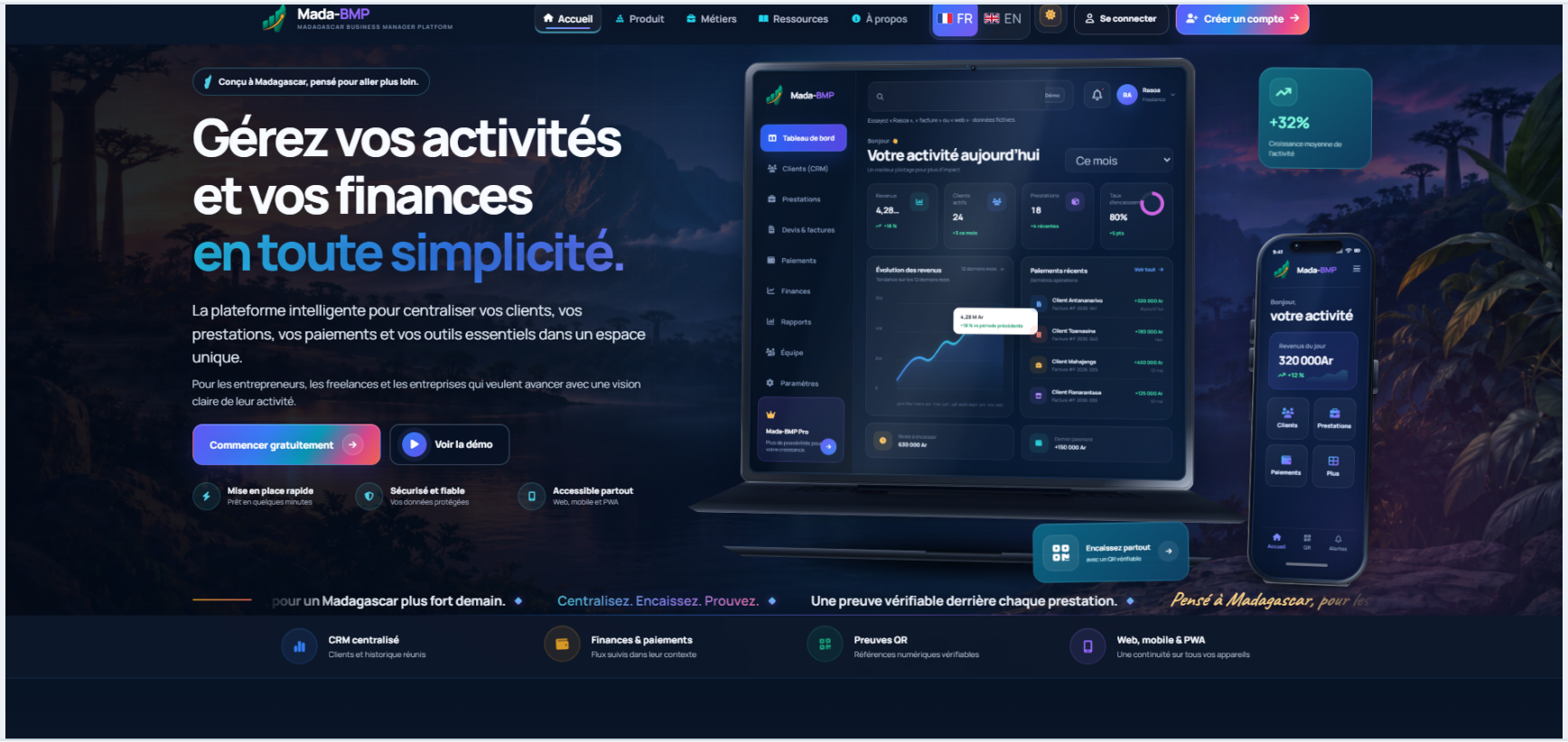
Freelancers, independent professionals, service providers, craftspeople and small businesses, depending on their actual needs.

TECHNOLOGIES · TARGET ARCHITECTURE

React / Vite / TypeScript frontend, Spring Boot backend, internal FastAPI / Python decision-intelligence service and PostgreSQL storage.

Visual presentation of Mada-BMP

Supplied public hero screenshot illustrating the project’s UI/UX direction.



Interface screenshot. Dashboard figures and indicators in the mock-up are fictional examples, not real commercial results.

Why design this platform?

The design starts from plausible real-world situations; its actual value still needs validation with users.

01 · FIND

Centralize the history of clients, requests and documents to avoid searching across several tools.

02 · TRACK

Clarify every stage of a job, from the initial request through payment and supporting documentation.

03 · COLLABORATE

Organize several team members, workspaces and permissions without sharing login credentials.

VALUE TO BE DEMONSTRATED · Fewer forgotten tasks, better continuity in customer follow-up and stronger traceability: benefits not yet measured in operation.

React frontend, Java backend and Python AI service

Spring orchestrates the internal FastAPI service described on the next page.

01 · USER INTERFACE

React / Vite / TypeScript. Components, forms and French/English public-facing journeys.

02 · COMMUNICATION

HTTP / JSON / REST API. Request/response contracts, validation and error handling.

03 · APPLICATION

Spring Boot / Security. Controllers, business services, sessions and authorizations.

04 · PERSISTENCE

PostgreSQL. Relational model, data integrity and workspace-scoped data.

SECURITY · Access rights must be checked on the server, and private data must be isolated across workspaces.

FastAPI: an analysis service behind Spring

Documented target flow; Spring remains responsible for business rules, sessions and authorization.

01 · FRONTEND

React / Vite / TypeScript

User interface, forms and presentation of analysis and recommendations.

→

02 · BUSINESS AUTHORITY

Spring Boot / Java 17

Authentication, workspace authorization, validation and orchestration.

→

03 · INTERNAL ANALYSIS

FastAPI / Python

Scores, summaries, analytical signals and explainable recommendations.

PERSISTENCE: POSTGRESQL 18

Business data and workspace isolation. FastAPI does not directly write sensitive business transactions.

TRUST BOUNDARY: Browser → Spring → FastAPI. AI cannot approve payments, refunds or role changes.

Support decisions, not replace them

Documented use cases; accuracy and end-to-end operation still require validation.

01 · CASH FLOW

Simple forecasts

Estimate cash-flow trends from available and validated data.

02 · BUDGET

Trends and anomalies

Flag potential overspending or unusual changes.

03 · FOLLOW-UP

Prioritization

Suggest which outstanding cases to review first, with reasoning.

04 · DASHBOARD

Useful summaries

Summarize indicators and highlight relevant signals for the user.

GOVERNANCE: minimize personal data, protect secrets, explain signals and provide a controlled fallback.

PROJECT STATUS · ONGOING MASTER 1 WORK

FastAPI and decision intelligence are part of the documented development scope. No trained model, accuracy score or commercial deployment is claimed without verified evidence.

Accounts, workspaces and distinct access rights

The model separates global identity, professional context, local role and business permissions.

IDENTITY · USER ACCOUNT

An authentication identity. A person can access multiple contexts according to the permissions granted.

CONTEXT · WORKSPACE

PERSONAL, FREELANCE or BUSINESS context; business data must remain isolated.

PERMISSIONS · AUTHORIZED ACCESS

OWNER, MANAGER, EMPLOYEE and dedicated permissions. A global ADMIN cannot freely read private business data.

DISTINCTION · The roles described belong to the design; their complete operational coverage must be verified.

From the initial request to evidence and history

A logical commercial workflow used to design user journeys and validation scenarios.

01 · CLIENT

Identify the person.

02 · REQUEST

Clarify the need.

03 · SERVICE

Track the work.

04 · QUOTATION

Propose a price.

05 · INVOICE

Issue a document.

06 · PAYMENT

Record the payment status.

07 · QR / EVIDENCE

Verify a reference.

BUSINESS RULE · Invoice, payment and receipt are separate objects; a QR code is not a money transfer.

Payment tracking, not a banking service

Financial scenarios in the project manual must be interpreted according to their actual nature.

DECLARATIVE ENTRY · CASH RECEIVED

The professional receives cash outside the application, then records the payment in the workspace.

EXTERNAL VERIFICATION · MOBILE MONEY

The professional checks receipt with the mobile operator and then records the transaction reference.

FUTURE DEVELOPMENT · INTEGRATED PAYMENT

A conditional evolution requiring an authorized payment provider, server-side confirmation and dedicated controls.

IMPORTANT LIMITATION · The balance calculated by the application is not a bank balance held or available for withdrawal.

Verification QR code: reference and status

The QR system represents documentary proof checked by the server, not a payment operation.

DOCUMENTED DESIGN · PURPOSE

Associate an identifier and a status with a receipt or document, then expose only the public information permitted in a verification response.

SECURITY REQUIREMENTS · EXPECTED CHECKS

Signature/integrity, expiry, document type, status, audit logging and separation of confidential data.

STATUS · The file describes a documented mechanism; it does not claim a complete production QR workflow.

Illustrative use cases, not completed sector products

The manual describes business examples to test the usefulness of the shared platform.

EXAMPLE 01 · COMPUTER REPAIR

Track a device, a diagnostic request, issued documents and the customer’s history.

EXAMPLE 02 · AGENCY & SMALL TEAM

Enable several people to work together with distinct responsibilities and protected data.

EXAMPLE 03 · SERVICES & APPOINTMENTS

Show service offerings and consider appointment requests as a possible business extension.

CAUTION · These scenarios do not establish the availability of 26 specialized modules or operational online booking.

Documented does not mean operational

An engineering approach distinguishes design decisions, demonstrations and real-world operation.

LEVEL 01 · DOCUMENTED

Architecture, rules and workflows described in reference material. This does not establish their complete execution.

LEVEL 02 · ILLUSTRATED / PLANNED

Mock-ups, fictional scenarios, selected business rules or development directions. Validation is still required.

LEVEL 03 · OPERATIONAL AND VERIFIED

End-to-end tested workflows, confirmed persistence, correctly enforced permissions and an identified runtime environment.

PUBLIC STATUS · Ongoing academic project: the documents do not establish commercial production, real revenue or measured adoption.

A core management platform before extensions

Validate the problem, the quality of operations and the tests before promising a complete platform.

P0 PRIORITY · CORE FOUNDATION

Accounts, clients, services, quotations, invoices, recorded payments and access control: the main scope to consolidate.

P1 EVOLUTION · PUBLIC CUSTOMER RELATIONS

Published services, requests, appointments and customer portal: possible evolutions to assess and validate.

NOT ESTABLISHED · FULL MARKETPLACE

Centralized collections, commissions, payouts or a stored-value wallet: outside the confirmed scope.

METHOD · Development and validation must precede any scope expansion or commercial claims.

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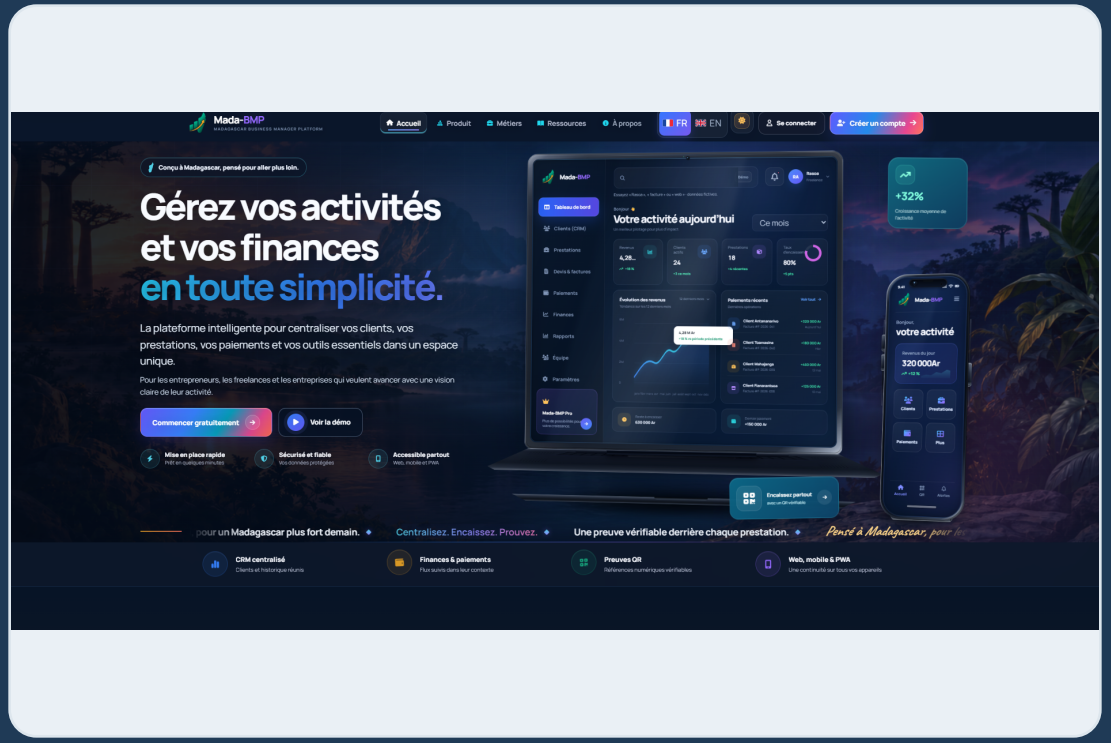
MADA-BMP

Master 1 academic project - in progress.

PROJECT OBJECTIVE

Design a business-management platform connecting clients, services, documents and payments, combining React, Spring Boot and PostgreSQL with an internal FastAPI decision-support service.

React / TypeScript · Spring Boot · PostgreSQL
FastAPI / Python · Decision intelligence



Illustrative public mock-up · fictional figures. No production deployment is claimed.

Presented by: RANDRIAMANANTENA Fitahiana Lafatra